

Mikko Numminen

Full-Stack Developer / AI-Assisted Workflows

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Portfolio: mikkonumminen.dev GitHub: github.com/MikkoNumminen



ABOUT

Full-stack developer working primarily with Next.js, TypeScript, and PostgreSQL. Currently maintain six production projects covering an HR system with role-based access control, a Turborepo-based community platform in active use, a browser game with a custom AI development pipeline, and a Windows audiobook generator. Strong focus on automated testing, security, and accessibility. Active use of Claude Code and multi-agent AI tooling in day-to-day work. Moved into software in 2022 after 24+ years in customer-facing senior sales roles.

SKILLS

Languages: TypeScript, JavaScript, Python 3.11+, SQL, Bash

Frontend: React 19, Next.js 15/16 (App Router), Astro, Three.js, GSAP, Phaser 3, Tailwind CSS, MUI 7

Backend: Next.js Server Actions, Prisma 6/7, Kysely, Zod, NextAuth v5, REST API integrations (Blizzard API, Open Library, Google Books)

Databases: PostgreSQL, Neon, MongoDB, raw SQL via typed query builders, dbmate

Testing: Jest, Vitest, Playwright E2E, React Testing Library, jest-axe (accessibility)

DevOps & CI: Docker, Kubernetes + Helm, GitHub Actions, Husky, Turborepo monorepos, Sentry, OpenTelemetry, PyInstaller + Inno Setup

Security & Compliance: RBAC with granular permissions, CSP with nonce, 2FA/TOTP, sliding-window rate limiting, audit logging, GDPR data export and deletion, WCAG accessibility

Cloud & Infra: Vercel, Cloudflare DNS, Azure (2 years prior employment)

PROJECTS

HRManager • Production-grade HR management system

Granular role-based access control, dashboard analytics, immutable audit trail in MongoDB, and AI-powered i18n across 18 languages.

- **1,828 tests at 91.9% line coverage:** 734 UI component tests, 253 server action tests, 75 Playwright E2E across RBAC, audit log, CSP, 2FA, and demo session isolation.
- **AI translation pipeline:** 18 locales kept in sync via parallel Claude Code agents (~80 s end-to-end) or headless Claude Haiku API for CI.
- **Demo session isolation:** each "Try Demo" click creates a private data sandbox, auto-expires after 24 h.

Stack: Next.js 16 • React 19 • TypeScript • MUI 7 + MUI X Charts • Prisma 7 • PostgreSQL + MongoDB • NextAuth v5 • Zod 4 • pg-boss • Sentry • OpenTelemetry • Jest 30 • Playwright • Docker • Vercel

hr-manager-pearl.vercel.app • github.com/MikkoNumminen/HRManager

Spacepotatis • Browser game with custom AI development pipeline

Single-player shooter with a 3D galaxy view and top-down combat, shipping with its own audio and AI-skill toolchain.

- **~1,170 Vitest tests** plus a documented incident response postmortem with mitigation-first methodology.
- **10 custom Claude Code skills**, version-controlled and quarterly drift-audited; estimated ~2.76M tokens saved annually across ~300 invocations.
- **Original audio toolchain:** music via Strudel live-coding (separate repo: strudel-patterns), voice via my AudiobookMaker on top of Chatterbox TTS. 100% copyright-free.

Stack: Next.js 15 • React 19 • Phaser 3 • Three.js • GSAP • Tailwind CSS • PostgreSQL • Kysely query builder • Neon • NextAuth • TypeScript strict • Vitest • Vercel

spacepotatis.vercel.app • github.com/MikkoNumminen/Spacepotatis

Platform (vuohiliitto.com) • Community platform in active production

Turborepo monorepo running a real online community with multi-tenant architecture, gamification system, and WoW-inspired tabbed chat.

- **1,388 tests across 153 suites + 42 Playwright E2E**, coverage threshold enforced (70% lines/functions) via pre-push hook.
- **Multi-tenant architecture**: two isolated services (Vuohiliitto + Platform) with 25 granular permission keys and zero-credential demo mode.
- **Gamification + WoW community features**: XP with 10 levels and 17 achievements, daily/weekly/custom quests, WoW-inspired chat with /w whispers, Blizzard API + Raider.IO M+ integration.
- **Production hardening**: GDPR compliance, 121-finding security audit (106 fixed), audit log of all admin actions, weekly soft-delete cron.

Stack: Next.js 15 • React 19 • TypeScript • MUI 7 • Prisma • PostgreSQL (Neon) • NextAuth v5 • Turborepo • Jest • Playwright • Husky • Vercel

vuohiliitto.com • github.com/MikkoNumminen/Platform

AudiobookMaker • Windows desktop app: PDF/EPUB to audiobook

Three pluggable TTS engines selectable in the GUI, shipped as a zero-dependency Windows installer with auto-updates from GitHub Releases.

- **2,172 tests** with pre-commit and pre-push enforcement covering TTS engines, normalizers, and audio pipeline.
- **18-pass Finnish text normalizer** with 300+ unit tests: handles numbers, dates, abbreviations, case inflection, loanwords.
- **Three TTS engines with subprocess isolation**: Edge-TTS online, Piper offline, Chatterbox GPU (isolated venv keeps heavy ML deps out of the main app); ~8× faster parallel CLI for batch books.

Stack: Python 3.11+ • PyMuPDF • edge-tts • piper-tts • Chatterbox • CustomTkinter • pydub + ffmpeg • PyInstaller • Inno Setup

github.com/MikkoNumminen/AudiobookMaker

mikkonumminen.dev • Personal portfolio site (itself a portfolio piece)

Hand-crafted counter-stack to my production work, used as the showcase entry point for the projects above.

- **Four crafted pages**: immersive WebGL scroll, interactive 3D solar system for projects, parallax mountain with climbing-goat scroll indicator, terminal/CRT contact page with a real command parser.
- **Multilingual (EN/FI/SV)**, comprehensive CSP + security headers, Sentry Core Web Vitals; fully static build, host-agnostic deploy (Vercel, S3+CloudFront, Cloudflare Pages).

Stack: Astro • Three.js • GSAP + ScrollTrigger • Tailwind CSS v4 • TypeScript strict • Sentry • Vercel

github.com/MikkoNumminen/mikkonumminen.dev

ReadLog • Book and audiobook reading tracker

Log books with format and finish date; search your library; "Have I read this?" lookup; public anonymous feed.

- **Parallel API integration** across Open Library and Google Books with result deduplication.
- **68 tests across 4 suites** covering API integrations, server actions, and components.
- **Google OAuth** sign-in, Neon Postgres, CI with type-check, lint, and full test suite on every push.

Stack: Next.js 16 • React 19 • TypeScript • Prisma 7 • PostgreSQL (Neon) • NextAuth v5 • MUI 7 • Jest • Open Library API • Google Books API

read-log.psi.vercel.app • github.com/MikkoNumminen/ReadLog

WORK EXPERIENCE

Kasvu Labs Oy • Software Developer 2022 – 2024

Full-stack development, data work, database management, UI design, and product maintenance.

Keijo Numminen Oy, Kesko Oyj, K-Niemi Oy • Senior Salesperson 1998 – 2022

24+ years in customer-facing senior sales roles: training and supervising associates, identifying customer needs, procurement, inventory management, and logistics for both private and professional customers.

EDUCATION & LANGUAGES

Tampere University • Computer Science / Tietotekniikka studies in progress

Finnish (native) • English (professional)